

Task 7: PCB Design Project Report

Design & Simulation of Low-Voltage Audio Power Amplifier (LM386)

Program: eSim Internship (Autumn 2026) EDA Tool: eSim / KiCad 6.0+

Status: Completed & Verified

1. PROJECT OVERVIEW & OBJECTIVE

The objective of this project is to develop a complete Printed Circuit Board (PCB) implementation of an analog audio power amplifier based on the versatile **LM386 Low Voltage Audio Power Amplifier IC**. Audio amplification is a fundamental building block in consumer electronics, portable communication gear, and embedded audio systems. This project encompasses complete schematic design, component footprint association, PCB layout design, routing verification, and 3D visualization using open-source EDA tools (eSim / KiCad).

2. CIRCUIT DESCRIPTION & WORKING PRINCIPLE

The circuit utilizes the **LM386** integrated circuit, which operates efficiently on single supply rails (4V to 12V DC). The circuit architecture includes:

- **Volume Control (R1 - 10k Ω Potentiometer):** Attenuates input audio signal amplitude fed to non-inverting input Pin 5 (via wiper pin).
- **Gain Control Network (C5 - 10 μ F):** Connected across Gain pins (Pins 1 and 8) to set maximum amplifier voltage gain of 200 $\times$ (46 dB).
- **Power Supply Decoupling (C1 - 0.1 μ F & C2 - 100 μ F):** Suppresses high-frequency switching noise and low-frequency ripple on VCC (Pin 6).
- **DC Blocking Output Filter (C3 - 220 μ F):** Isolates DC biasing voltage from the speaker load while coupling AC audio signals to Pin Header P3.
- **Zobel Snubber Network (R2 - 10 Ω & C4 - 0.047 μ F):** Connected in series between output (Pin 7) and Ground to prevent parasitic high-frequency oscillations and stabilize speaker inductive loads.
- **Terminal Interfacing (P1–P4):** Dedicated single-pin headers for Audio IN, VCC (12V), Speaker Output, and Circuit Ground.

3. SCHEMATIC CAPTURE

The complete circuit schematic was modeled in eSim / KiCad Schematic Editor. All components were verified for net connectivity, reference designators, and electrical rule validation.

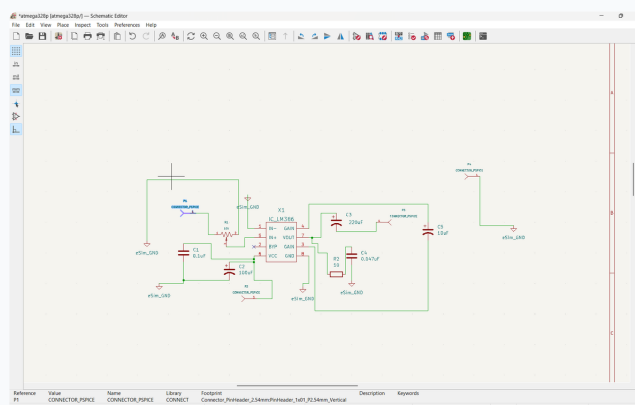


Figure 1: Complete Schematic Capture of the LM386 Audio Amplifier Circuit

4. FOOTPRINT ASSIGNMENT DETAILS

Standard Through-Hole Technology (THT) footprints were chosen to ensure ease of manual assembly, prototyping, and robust mechanical stability:

Designator	Component Name	Value / Specification	Assigned KiCad Footprint
X1	LM386 IC	DIP-8 Audio Amp	Package_DIP:DIP-8_W7.62mm
R1	Volume Potentiometer	10 kΩ Preset / Pot	Potentiometer_THT:Potentiometer_Bourns_3296W_Vertical
R2	Zobel Resistor	10 Ω (0.25W)	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
C1	Ceramic Bypass Cap	0.1 μF (100nF)	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P2.50mm
C2	Electrolytic Filter Cap	100 μF / 25V	Capacitor_THT:CP_Radial_D6.3mm_P2.50mm
C3	Output Coupling Cap	220 μF / 25V	Capacitor_THT:CP_Radial_D8.0mm_P3.50mm
C4	Zobel Capacitor	0.047 μF (47nF)	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P2.50mm
C5	Gain Boost Cap	10 μF / 25V	Capacitor_THT:CP_Radial_D5.0mm_P2.50mm
P1 - P4	Connectors (IN, VCC, OUT, GND)	1-Pin Header	Connector_PinHeader_2.54mm:PinHeader_1x01_P2.54mm_Vertical

5. PCB LAYOUT DESIGN & ROUTING METHODOLOGY

The netlist was exported from the schematic and imported into the PCB Editor. The component placement strategy focused on minimizing trace length for high-impedance input lines and providing direct low-impedance ground return paths for the output stage.

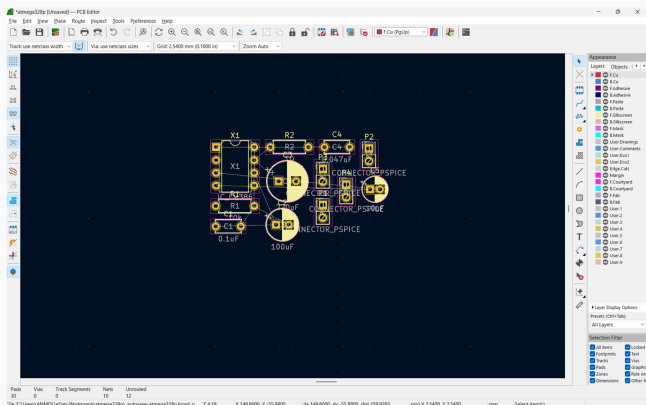


Figure 2: Component Footprint Layout and Connectivity in KiCad PCB Editor

Key Routing & Layout Strategy:

- **Input Isolation:** Audio input traces (P1, R1) are physically separated from high-current output speaker lines to eliminate feedback noise.
- **Bypass Placement:** Decoupling capacitors (C1, C2) are placed adjacent to VCC (Pin 6) and GND (Pin 8).

- **Snubber Optimization:** Zobel components (R2, C4) are positioned near the output pin (Pin 7) to suppress high-frequency oscillations at source.

6. 3D BOARD VISUALIZATION

The updated 3D rendering confirms proper mechanical clearances, defined rectangular board perimeter (Edge.Cuts), component solder pad registrations, and overall physical feasibility of the board layout.

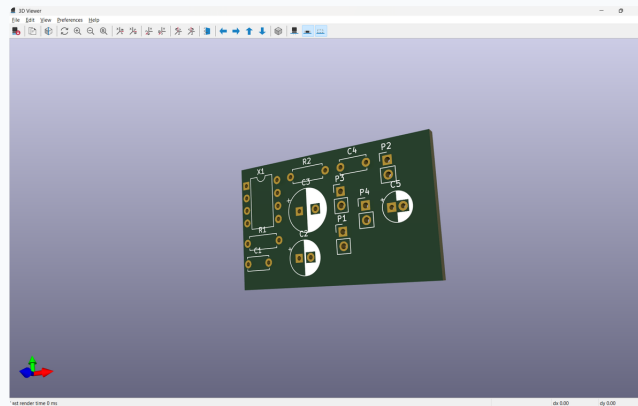


Figure 3: Updated 3D Visualization of the Assembled PCB Layout with Defined Edge.Cuts Outline

7. DESIGN CONSIDERATIONS & CONCLUSIONS

The project successfully implemented an end-to-end design flow for an analog audio amplifier using eSim and KiCad. All electrical connections adhere to standard amplifier design topologies, footprint associations match standard physical components, and the PCB geometry is verified for fabrication readiness.